LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Generator, engine coupling, cooling of generators, constructional details of brushes types of brushes and causes of brush wear and spark suppression
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Generator, engine coupling, cooling of generators, constructional details of brushes types of brushes and causes of brush wear and spark suppression techniques.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the A/C power systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of brush assembly removal and inspection; Safety: Ensure power is isolated and capacitors discharged; Supervision: Instructor monitors tool usage and electrical safety protocols.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft AC power system?
(b) How does the engine drive the generator mechanically?
(c) Why is spark suppression critical in generator maintenance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Generator engine coupling mechanisms • Generator cooling systems and airflow • Constructional details of generator stator and rotor • Types of brushes and commutator materials • Causes of brush wear and replacement criteria	L & Ds	PPT, IB	15 min
(b)	• Spark suppression techniques and components • Brush wear analysis and fault diagnosis • Inspection procedures for generator coupling • Cooling system maintenance checks • Brush material selection and application	L & Ds	PPT, PDF	20 min
(c)	• Review of AC power system integration • Summary of brush wear causes and prevention • Overview of spark suppression methods • Key constructional features of generators • Cooling system operational requirements	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between brush wear and commutator sparking.
(b) Describe the role of the carbon pile in voltage regulation context.
(c) Identify three common causes of excessive generator brush wear.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between carbon pile and solid state voltage regulators to be discussed in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Need for Voltage regulation, types of voltage Regulators - Carbon pile voltage regulator, solid voltage regulators, paralleling and load sharing.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.
6. Verify specific BR reference number for brush wear charts.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min